

Winding-Free Exact Diagonalization of Quantum Spin Ice

Zhengbang Zhou^{1,*} and Yong Baek Kim^{1,†}

¹*Department of Physics, University of Toronto, Toronto, Ontario M5S 1A7, Canada*

(Dated: July 22, 2026)

Quantum spin ice realizes an emergent three-dimensional $U(1)$ gauge theory whose low-energy dynamics is generated by six-spin ring exchange around pyrochlore hexagons. The sign of this exchange selects zero- or π -flux gauge backgrounds, yet the frustrated sign relevant to π -flux candidates is inaccessible to sign-problem-free quantum Monte Carlo. Exact diagonalization is sign unrestricted, but standard periodic clusters admit a noncontractible four-spin process at second order, parametrically above the physical third-order hexagon tunneling. We eliminate this finite-torus artifact by coupling microscopic spin transfers to a source that records their transport on the covering lattice and projecting the exact ice-descended band onto its zero-transport Fourier component. On cubic-16, the $M = 3$ and $M = 4$ projected operators agree within 0.013% in centered operator norm, and a third-order path classification removes every winding four-loop while retaining every contractible hexagon. At $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, the low-temperature heat-capacity peak shifts from $0.02554J_{zz}$ in ordinary exact diagonalization to $0.001463J_{zz}$, close to the quantum Monte Carlo value $0.001100J_{zz}$. The longitudinal X -point excitation centroid simultaneously shifts from $0.08069J_{zz}$ to $0.005362J_{zz}$. The residual thermodynamic discrepancies isolate finite-size effects beyond the removed winding process.

Introduction.—Quantum spin ice (QSI) is a $U(1)$ quantum spin liquid with fractionalized excitations and an emergent gauge field whose gapless photon mode provides a characteristic low-energy signature [1–3]. Recent experiments on $\text{Ce}_2(\text{Zr}, \text{Sn}, \text{Hf})_2\text{O}_7$ suggest that Ce-based pyrochlores may realize a π -flux QSI, in which an emergent gauge flux of π threads each elementary hexagonal plaquette [Fig. 1(i.a)] [4–6]. In $\text{Ce}_2\text{Zr}_2\text{O}_7$, no long-range order or spin freezing is observed down to the lowest accessible temperatures, whereas recent heat-capacity measurements on $\text{Ce}_2\text{Hf}_2\text{O}_7$ resolve two anomalies near 65 mK and 25 mK [7]. Numerical linked-cluster fits place the candidate parameters within the π -flux QSI regime, motivating a microscopic explanation of this two-scale thermodynamic structure.

The minimal microscopic setting for QSI is the nearest-neighbor pseudospin-1/2 XYZ model

$$H_0 = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-), \quad (1)$$

with $J_{zz} > 0$. For $|J_{\pm}|, |J_{\pm\pm}| \ll J_{zz}$ the low-energy states obey the spin-ice rule $Q_t \equiv \sum_{i \in t} S_i^z = 0$ on every tetrahedron t ; we write P_{ice} for the orthogonal projector onto that manifold, and call a tetrahedron with $Q_t \neq 0$ a spinon. Transverse exchange first connects distinct ice configurations at third order, through collective flips around elementary six-site hexagons, generating the ring-exchange scale

$$g_6 = \frac{12|J_{\pm}|^3}{J_{zz}^2}. \quad (2)$$

In the convention of Eq. (1), $J_{\pm} > 0$ gives the zero-flux QSI, whereas $J_{\pm} < 0$ gives the π -flux QSI. The paramet-

rically small scale g_6 is therefore a natural candidate for the lower-temperature thermodynamic anomaly.

Obtaining a microscopic description of this regime using unbiased numerical methods remains challenging. Quantum Monte Carlo (QMC) simulations access the zero-flux QSI for $J_{\pm} > 0$ [8–10] but encounter a sign problem in the π -flux regime, $J_{\pm} < 0$. Numerical linked-cluster expansions fail to converge at the ultralow temperatures at which the lower anomaly emerges. Finite-temperature tensor-network calculations have reached a periodic 48-site three-dimensional Heisenberg cluster, but only well above the ring-exchange scale [11]. Density-matrix renormalization group and minimally entangled typical thermal states reach lower temperatures on pyrochlore tubes [12], but their quasi-one-dimensional geometry cannot support a gapless photon mode. A sign-unrestricted microscopic treatment that resolves the three-dimensional gauge dynamics at temperatures of order g_6 is therefore needed.

Exact diagonalization (ED) is exact on a finite cluster and imposes no sign restriction, but is limited to small pyrochlore tori, notably the cubic 16-site and face-centered-cubic 32-site clusters. Their dominant artifact is not size alone but the appearance of spurious four-site loops. On cubic-16, a noncontractible four-spin path closes through the periodic boundary [Figs. 1(i.b) and 1(i.c)] and contributes already at second order, with

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad \frac{g_4}{g_6} = \frac{J_{zz}}{3|J_{\pm}|}, \quad (3)$$

whereas the physical hexagon process appears only at third order, at the scale g_6 . The artifact therefore diverges relative to the physical process in the nominally controlled limit $|J_{\pm}|/J_{zz} \rightarrow 0$. Cubic-16 hosts 36 such winding four-cycles and 16 contractible hexagons; FCC-32 reduces but does not remove the problem, with 24

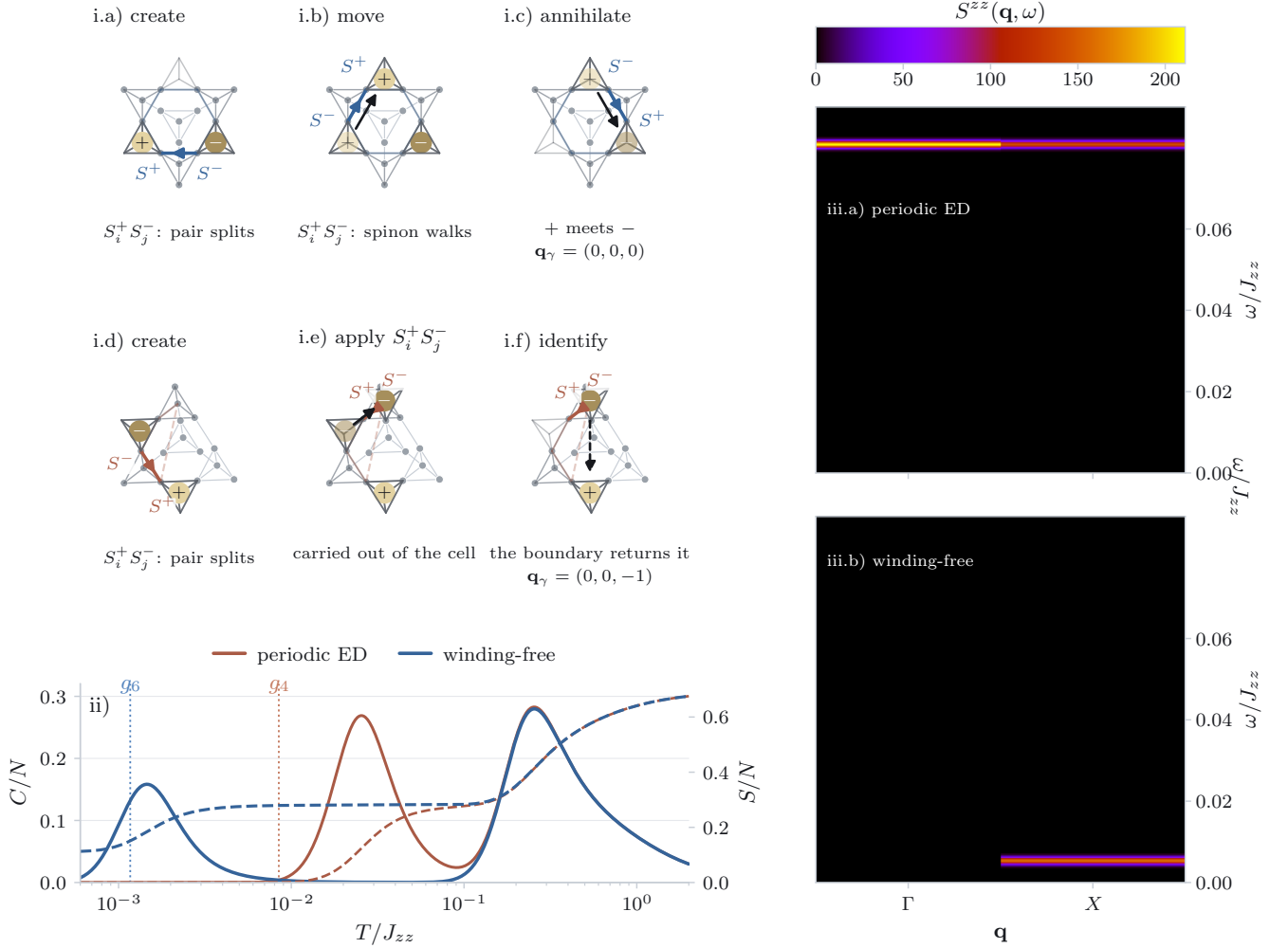


FIG. 1. Microscopic transport and winding removal on cubic-16. (i) A ring exchange stage by stage on the 16-site torus; gray marks the cluster, colored circles the loop, filled versus open symbols the alternating S^z pattern the flip reverses. Each stage highlights the bond its exchange acts on, its endpoints labeled $S^\pm$ by which spin is raised and which lowered. The move leaves the tetrahedron it straddles neutral and charges the two outer ones, so it creates, moves, or annihilates a spinon–antispinon pair. Charges sit at the center of their tetrahedron, whose frame is drawn, light for $Q_t > 0$ and dark for $Q_t < 0$; an arrow marks every transfer. (i.a)–(i.c) The contractible hexagon: one exchange splits a pair, the next walks the spinon on, the third brings $+$ into $-$. All inside one cell, and nothing is transported, $\mathbf{q}_\gamma = \mathbf{0}$. (i.d)–(i.f) The axial four-loop, with its closing step split into the two operations it is actually made of. (i.d) The first exchange splits a pair, exactly as on the hexagon. (i.e) The second exchange then acts, and the antispinon stays put while the spinon is carried out of the cell entirely: the bond it travels along closes only through the periodic boundary, so on the covering lattice the two charges end a full period apart and have not annihilated. (i.f) Only the periodic identification, drawn dashed, maps the spinon back onto its partner and closes the pair. That second operation has no counterpart on the infinite lattice, which is what $\mathbf{q}_\gamma = (0, 0, -1)$ records: the sequence lifts to an open path. The diagonal four-loop behaves identically with $\mathbf{q}_\gamma = (1, 1, 0)$. The source average of Eq. (10) keeps the hexagon and removes both four-loops. (ii) Heat capacity (solid, left axis) and entropy (dashed, right axis) at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$, for ordinary periodic ED (orange) and the winding-free replacement of Eq. (11) at $M = 4$ (blue). Dotted vertical lines mark the perturbative scales g_6 and g_4 of Eqs. (2) and (3); the periodic low-temperature peak tracks the artifact scale g_4 , whereas the projected peak moves down to the ring-exchange scale g_6 . (iii.a),(iii.b) Zero-temperature longitudinal structure factor $S^{zz}(\mathbf{k}, \omega)$ [Eq. (15)] for the periodic band $h_{M=4}$ and the projected band $h_{M=4}$, on identical linear color scales. Both use the same canonically dressed microscopic S^z probe [Eq. (14)]; results are averaged over the three symmetry-related X points, with broadening $\eta = 6 \times 10^{-4} J_{zz}$.

winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons. Ordinary ED thus probes the parametrically larger scale of a finite-torus winding process

rather than the ring exchange governing the emergent gauge sector, and the lower anomaly it produces is centered near g_4 rather than g_6 , as shown in Fig. 1(ii).

In this Letter, we remove this artifact by coupling microscopic spin transfers to an auxiliary source that records their transport on the covering lattice. We map the exact ice-descended band to a fixed reference space and retain only its zero-transport Fourier component. This transport-character projection removes noncontractible winding processes while preserving contractible hexagon dynamics and microscopic virtual dressing. On cubic-16, the projected band is converged to sub-percent accuracy with source-grid size and recovers the correct low-energy thermodynamic scale at a sign-free benchmark point. The projection also strongly reorganizes the longitudinal spectrum, demonstrating that the winding artifact affects both energies and matrix elements.

Transport-character projection.—The physical and spurious processes can be separated without reference to perturbative order. A microscopic sequence may return to the same spin configuration on the finite torus while transporting Ising polarization through a periodic boundary: it is closed on the torus, but fails to close when lifted continuously to the infinite covering lattice. The winding four-cycles transport nonzero polarization, whereas a contractible hexagon transports none [Figs. 1(i.a)–1(i.f)]. Equivalently, each exchange creates or moves a spinon–antispinon pair, and the sequence returns to the ice manifold only by annihilating it: on a contractible loop the two charges genuinely meet, whereas on a winding loop each meets a periodic image of its partner, which is why the process survives on the torus but has no counterpart on the covering lattice. We now promote this observation to an exact operator identity.

A periodic cluster is specified by three period vectors, collected as the rows of a 3×3 matrix L , and each site i is assigned a representative position $\mathbf{r}_i$ in a chosen fundamental cell. For every stored oriented nearest-neighbor bond (i, j) we choose $\mathbf{n}_{ij} \in \mathbb{Z}^3$ such that the covering-lattice image of j connected directly to i lies at $\mathbf{R}_{j|i} = \mathbf{r}_j - \mathbf{n}_{ij}^T L$; thus $\mathbf{n}_{ij} = 0$ for an intracell bond, whereas $\mathbf{n}_{ij} \neq 0$ records the periodic image reached by a boundary-crossing bond. The covering-lattice displacement along the oriented bond and the corresponding pair coordinate are $\mathbf{d}_{ij} = \mathbf{R}_{j|i} - \mathbf{r}_i$ and $\mathbf{c}_{ij} = \mathbf{r}_i + \mathbf{R}_{j|i}$.

To quantify transport, consider a spin configuration $\mathcal{C}$ lifted from the torus to the covering lattice, and associate with it the polarization coordinate

$$\mathbf{p}(\mathcal{C}) = \sum_{i=1}^N S_i^z(\mathcal{C}) \mathbf{R}_i, \quad (4)$$

where i runs over all N cluster sites and $\mathbf{R}_i$ is the covering-lattice position assigned to site i by the lift. Equation (4) is the first spatial moment of the site-resolved Ising polarization. It introduces no new physical degree of freedom and is neither an electric nor a magnetic dipole moment, nor a dipole formed from the tetrahedral charges Q_t ; it is a bookkeeping coordinate

recording how S^z is displaced by a sequence of microscopic spin flips.

Taken alone, $\mathbf{p}$ is not single valued on a torus: replacing a site by one of its periodic images shifts it by a period vector. Its change along a continuously lifted sequence is unambiguous, however, and for an ice-to-ice sequence is also independent of the coordinate origin, because the ice rule implies $\sum_i S_i^z = 0$. The increments then follow directly from Eq. (4): $S_i^+ S_j^-$ moves one unit of S^z from $\mathbf{R}_{j|i}$ to $\mathbf{r}_i$, giving $\Delta \mathbf{p} = -\mathbf{d}_{ij}$, whereas $S_i^+ S_j^+$ raises both endpoints and displaces $\mathbf{p}$ by their summed position $\mathbf{c}_{ij}$. A move T_ℓ on the oriented bond (i, j) therefore contributes

$$\Delta \mathbf{p}_\ell = \begin{cases} \mp \mathbf{d}_{ij}, & T_\ell = S_i^\pm S_j^\mp, \\ \pm \mathbf{c}_{ij}, & T_\ell = S_i^\pm S_j^\pm. \end{cases} \quad (5)$$

We introduce an auxiliary counting field $\boldsymbol{\theta}$ conjugate to these increments by multiplying each microscopic move by $e^{-2i\boldsymbol{\theta} \cdot \Delta \mathbf{p}_\ell}$, giving the sourced Hamiltonian

$$H(\boldsymbol{\theta}) = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij \rangle} (e^{2i\boldsymbol{\theta} \cdot \mathbf{d}_{ij}} S_i^+ S_j^- + \text{h.c.}) + J_{\pm\pm} \sum_{\langle ij \rangle} (e^{-2i\boldsymbol{\theta} \cdot \mathbf{c}_{ij}} S_i^+ S_j^+ + \text{h.c.}). \quad (6)$$

The field $\boldsymbol{\theta}$ is not a physical electromagnetic field; it is a Fourier variable that records the polarization transported by microscopic spin transfers. For any sequence γ the source phases multiply to exactly $e^{-i\boldsymbol{\theta} \cdot \mathbf{q}_\gamma}$, with

$$\mathbf{q}_\gamma = 2 \sum_{\ell \in \gamma} \Delta \mathbf{p}_\ell, \quad (7)$$

the factor of two expressing the transport in the natural normalization $\sigma^z = 2S^z = \pm 1$. The vector $\mathbf{q}_\gamma$ is thus the net Ising-polarization transport of the lifted sequence.

Crucially, $\mathbf{q}_\gamma$ is a property of the path through configuration space, not merely of its endpoints on the finite torus. For a contractible closed process the lifted configuration also closes, the polarization returns to the same branch, and $\mathbf{q}_\gamma = 0$. A winding process instead returns to precisely the same finite-cluster configuration while transferring polarization through a periodic boundary; its lift ends on a different branch and $\mathbf{q}_\gamma \neq 0$. Equation (7) is therefore an exact microscopic winding label, valid beyond perturbation theory and in the presence of both $J_\pm$ and $J_{\pm\pm}$.

Let P_θ be the spectral projector onto the isolated band adiabatically connected to the spin-ice manifold,

$$P_\theta = \frac{1}{2\pi i} \oint_{\Gamma_\theta} (z - H(\boldsymbol{\theta}))^{-1} dz, \quad \text{rank } P_\theta = \text{rank } P_{\text{ice}}, \quad (8)$$

where the contour Γ_θ encloses the selected band and no other part of the spectrum. Because the band eigenvectors depend on $\boldsymbol{\theta}$, their matrix representations cannot be

averaged directly. We therefore identify every sourced band with the same fixed ice-rule reference space using the polar isometry W_{θ} , and define the exact band Hamiltonian in that common basis,

$$W_{\theta} = P_{\theta} P_{\text{ice}} (P_{\text{ice}} P_{\theta} P_{\text{ice}})^{-1/2}, \quad h(\theta) = W_{\theta}^{\dagger} H(\theta) W_{\theta}. \quad (9)$$

A continuous source average retains only the zero-transport Fourier component, which we evaluate numerically on an M^3 grid,

$$h_{\text{wf}} = \frac{1}{(2\pi)^3} \int_{[0,2\pi]^3} d^3\theta h(\theta) \simeq h_M = \frac{1}{M^3} \sum_{\mathbf{m} \in \mathbb{Z}_M^3} h\left(\frac{2\pi\mathbf{m}}{M}\right). \quad (10)$$

The continuous average selects $\mathbf{q}_{\gamma} = 0$ exactly, removing every process that carries nonzero covering-lattice transport; the discrete grid instead retains all characters with $\mathbf{q}_{\gamma} = \mathbf{0} \pmod{M}$, so convergence in M is an essential diagnostic. One consequence of the continuous average is available exactly at any M : since $\mathbf{q}_{\gamma} = 2\Delta\mathbf{p}$, a surviving matrix element cannot change $\mathbf{p}$, so h_{wf} is block diagonal in the polarization coordinate of Eq. (4). We therefore impose that block structure directly on h_M . The residual inter-sector coupling it discards is only $1.6 \times 10^{-5} J_{zz}$ at $M = 4$, but it is comparable to the splittings within the low-lying manifold and would otherwise lift degeneracies that the exact projection protects. The construction ceases to be well defined if the selected band closes its spectral gap or if $P_{\text{ice}} P_{\theta} P_{\text{ice}}$ becomes singular.

For thermodynamics we replace only the exact band descended from the ice manifold,

$$Z_{\text{wf}}(\beta) = Z_{\text{full}}(\beta) - \text{Tr}_{\text{ice}} e^{-\beta h(\mathbf{0})} + \text{Tr}_{\text{ice}} e^{-\beta \bar{h}_M}, \quad (11)$$

and perform the analogous substitution for the first two energy moments that give the internal energy and heat capacity; the microscopic spectrum outside the selected band is left unchanged. Here

$$\bar{h}_M = h_M + \frac{\text{Tr}[h(\mathbf{0}) - h_M]}{d} I, \quad d = \text{rank } P_{\text{ice}}, \quad (12)$$

fixes the scalar placement of the corrected band relative to the microscopic spectrum. The band Hamiltonians of Eq. (9) are referred to the ice manifold and therefore carry no absolute energy origin, which the character average of Eq. (10) does not supply; matching the band trace to $h(\mathbf{0})$ supplies it. Because a uniform shift of an isolated band cancels in both C and S , this choice leaves the reported low-temperature observables unchanged. Equation (11) defines a finite-size diagnostic rather than the partition function of the original periodic Hamiltonian. Derivations and numerical details are given in the Supplemental Material [13].

Topology and convergence.—We first test the transport label by explicitly enumerating all ice-to-ice sequences through third order. On both cubic-16 and FCC-32, the

zero-character projection removes every winding four-cycle and wrapping hexagon while retaining every contractible hexagon. This perturbative enumeration serves only to verify the topological classification; the projected band itself is constructed nonperturbatively from the exact eigenvalues of $H(\theta)$. At the sign-free benchmark point

$$(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0), \quad (13)$$

this third-order path diagnostic shifts the cubic-16 heat-capacity maximum from $T_p/J_{zz} = 0.02554$ to 0.001252, and the FCC-32 maximum from 0.008770 to 0.000608. The FCC-32 values follow from the path-based effective description rather than from the complete finite-temperature spectrum.

For cubic-16, the exact projected band is well resolved throughout the source grid. At $\theta = \mathbf{0}$ the smallest overlap between the selected band and the ice manifold is 0.762, and the gap to the remaining spectrum in the working total- S^z sector is $0.627 J_{zz}$; every point on the $M = 3$ and $M = 4$ grids has both a larger overlap and a larger gap, with a largest Ritz residual of 3.7×10^{-12} . The band is also the lowest $d = 90$ levels of the complete 65 536-state spectrum, separated from the rest of it by $0.3874 J_{zz}$; this is the gap that isolates the replaced band in Eq. (11), and it exceeds the low-temperature peak scale by more than two orders of magnitude. After subtracting the scalar component of the band Hamiltonian, the relative change is 6.36% between $M = 2$ and $M = 3$, but only 0.013% between $M = 3$ and $M = 4$. The larger discrepancy at $M = 2$ is character aliasing: although the $M = 2$ grid removes the shortest winding four-cycles, it folds every transport sector with $\mathbf{q}_{\gamma} = 2\ell$, $\ell \in \mathbb{Z}^3$, into the zero character. Increasing M separates these aliased sectors.

Thermodynamics and dynamics.—The sign-free point of Eq. (13) provides an external thermodynamic benchmark. Continuous-time worm-QMC results for Eq. (1) are available on a lattice of 8^3 primitive cells [10]; digitizing the published curve gives heat-capacity maxima at $T/J_{zz} \approx 0.001100$ and 0.2105. Ordinary cubic-16 ED instead places the low-temperature maximum at $T/J_{zz} = 0.02554$, where it is controlled by the noncontractible four-loop scale, whereas the converged $M = 4$ replacement shifts it to $T/J_{zz} = 0.001463$ [Fig. 1(ii)]. The projected result is therefore close to both the physical hexagon scale and the QMC maximum. Over the low-temperature comparison window the heat-capacity root-mean-square error decreases from 0.0859 to 0.02327, while the entropy error remains 0.04509: removing the winding process corrects the dominant low-energy scale without making cubic-16 a quantitatively accurate representation of the bulk system.

We next examine the effect of the projection on the longitudinal dynamical response. To use the same microscopic probe in the periodic and projected bands, we

map the spin operator into the fixed ice reference space with the zero-source isometry W_0 , defining for sublattice a

$$\tilde{S}_a^z(\mathbf{k}) = W_0^\dagger \left(\sum_{j \in a} e^{-2\pi i \mathbf{k} \cdot \mathbf{r}_j} S_j^z \right) W_0, \quad (14)$$

where $\mathbf{k}$ is the reciprocal-space momentum, distinct from the transport label $\mathbf{q}_\gamma$. For either the periodic band Hamiltonian $h(\mathbf{0})$ or the projected Hamiltonian $h_{M=4}$, the zero-temperature longitudinal structure factor is

$$S^{zz}(\mathbf{k}, \omega) = \frac{1}{Nd_0} \sum_{g \in \mathcal{G}} \sum_{n \notin \mathcal{G}} \sum_{a=0}^3 \left| \langle n | \tilde{S}_a^z(\mathbf{k}) | g \rangle \right|^2 \times G_\eta [\omega - (E_n - E_g)], \quad (15)$$

where $\mathcal{G}$ is the ground-state manifold, $d_0 = |\mathcal{G}|$, N is the number of sites, $\eta = 6 \times 10^{-4} J_{zz}$ is the spectral broadening, and $|g\rangle$, $|n\rangle$ are eigenstates of the band Hamiltonian under consideration.

Cubic-16 resolves only the Γ point and three symmetry-related X points. As an internal consistency check, the operator obtained from the zero-source isometry agrees with an independently constructed periodic-band operator to a relative error of 2.8×10^{-14} . The periodic band has a unique ground state, whereas the projected band has a sixfold one: the six polarization sectors $\mathbf{p} = (\pm\frac{1}{2}, 0, 0)$, $(0, \pm\frac{1}{2}, 0)$, $(0, 0, \pm\frac{1}{2})$ can no longer tunnel into one another and stay exactly degenerate. At X , the spectral centroid shifts from $0.08069 J_{zz}$ in the periodic band to $0.005362 J_{zz}$ after projection [Figs. 1(iii.a) and 1(iii.b)]. At Γ the projection instead removes the response almost entirely, suppressing the integrated weight from 0.3182 to 2.1×10^{-6} : the sublattice magnetization commutes with $\mathbf{p}$, so once the sectors are decoupled it can no longer carry the ground manifold outside itself. The projection therefore reorganizes both the excitation energies and their spectral weights, rather than merely rescaling the periodic spectrum.

That sixfold degeneracy leaves a residual entropy $\ln 6$ per cluster, or 0.1120 per site, which the winding-free band never releases: the projection makes $\mathbf{p}$ exactly conserved, so the sectors cannot mix at any temperature. The count of such sectors grows only polynomially with linear size, so this contribution is subextensive and vanishes per site in the thermodynamic limit, but on cubic-16 it is a floor under the low-temperature entropy and accounts for part of the residual discrepancy with QMC.

Discussion.—Periodic ED can replace the physical hexagon dynamics of QSI with a parametrically larger noncontractible four-spin process. Transport-character projection removes this winding contribution from the exact ice-descended band while preserving contractible processes, microscopic virtual dressing, and the spectrum outside the selected band. It is supported by the exact source-phase identity, the exhaustive classification of

short ice-to-ice paths, and agreement to 0.013% between the $M = 3$ and $M = 4$ grids.

The QMC comparison also clarifies what the projection does not correct. Cubic-16 has no long-wavelength photon sector and only sparse momentum resolution, so its thermodynamics and dynamical response remain size dependent even after the winding scale is removed: the projection is a controlled finite-torus correction, not a replacement for thermodynamic extrapolation.

A character-converged FCC-32 calculation at the full XYZ parameters relevant to $\text{Ce}_2\text{Hf}_2\text{O}_7$ is the natural next step. There, with sign-free QMC unavailable, reliability must be assessed through consistency between cluster geometries and across thermodynamic, entropic, and neutron observables. More broadly, transport-character projection provides a route for extracting local gauge dynamics from finite periodic clusters without allowing the topology of the simulation torus to determine the leading low-energy process.

* zhengbang.zhou@mail.utoronto.ca

† ybkim@physics.utoronto.ca

- [1] M. Hermele, M. P. A. Fisher, and L. Balents, Pyrochlore photons: The $U(1)$ spin liquid in a $S = 1/2$ three-dimensional frustrated magnet, *Phys. Rev. B* **69**, 064404 (2004).
- [2] M. J. P. Gingras and P. A. McClarty, Quantum spin ice: A search for gapless quantum spin liquids in pyrochlore magnets, *Rep. Prog. Phys.* **77**, 056501 (2014).
- [3] O. Benton, O. Sikora, and N. Shannon, Seeing the light: Experimental signatures of emergent electromagnetism in a quantum spin ice, *Phys. Rev. B* **86**, 075154 (2012).
- [4] B. Gao, T. Chen, D. W. Tam, C.-L. Huang, K. Sasmal, D. T. Adroja, F. Ye, H. Cao, G. Sala, M. B. Stone, C. Baines, J. A. T. Barker, W. Hu, J.-H. Chung, X. Xu, S.-W. Cheong, M. Nallaiyan, S. Spagna, M. B. Maple, A. H. Nevidomskyy, E. Morosan, G. Chen, and P. Dai, Experimental signatures of a three-dimensional quantum spin liquid in effective spin-1/2 $\text{Ce}_2\text{Zr}_2\text{O}_7$ pyrochlore, *Nat. Phys.* **15**, 1052 (2019).
- [5] J. Gaudet, E. M. Smith, J. Dudemaine, J. Beare, C. R. C. Buhariwalla, N. P. Butch, M. B. Stone, A. I. Kolesnikov, G. Xu, D. R. Yahne, K. A. Ross, C. A. Marjerrison, J. D. Garrett, G. M. Luke, A. D. Bianchi, and B. D. Gaulin, Quantum spin ice dynamics in the dipole-octupole pyrochlore magnet $\text{Ce}_2\text{Zr}_2\text{O}_7$, *Phys. Rev. Lett.* **122**, 187201 (2019).
- [6] A. Bhardwaj, S. Zhang, H. Yan, R. Moessner, A. H. Nevidomskyy, and H. J. Changlani, Sleuthing out exotic quantum spin liquidity in the pyrochlore magnet $\text{Ce}_2\text{Zr}_2\text{O}_7$, *npj Quantum Mater.* **7**, 51 (2022).
- [7] E. M. Smith, A. Fitterman, R. Sch" afer, *et al.*, Two-peak heat capacity accounts for $R \ln 2$ entropy and ground state access in the dipole-octupole pyrochlore $\text{Ce}_2\text{Hf}_2\text{O}_7$, *Phys. Rev. Lett.* **135**, 086702 (2025).
- [8] N. Shannon, O. Sikora, F. Pollmann, K. Penc, and P. Fulde, Quantum ice: A quantum monte carlo study, *Phys. Rev. Lett.* **108**, 067204 (2012).

- [9] Y. Kato and S. Onoda, Numerical evidence of quantum melting of spin ice: Quantum-to-classical crossover, *Phys. Rev. Lett.* **115**, 077202 (2015).
- [10] C.-J. Huang, Y. Deng, Y. Wan, and Z. Y. Meng, Dynamics of topological excitations in a model quantum spin ice, *Phys. Rev. Lett.* **120**, 167202 (2018), [arXiv:1707.00099](#).
- [11] R. Schäfer, I. Hagymási, R. Moessner, and D. J. Luitz, Pyrochlore $S = \frac{1}{2}$ heisenberg antiferromagnet at finite temperature, *Phys. Rev. B* **102**, 054408 (2020).
- [12] I. Hagymási, R. Schäfer, R. Moessner, and D. J. Luitz, Possible inversion symmetry breaking in the $S = 1/2$ pyrochlore heisenberg magnet, *Phys. Rev. Lett.* **126**, 117204 (2021).
- [13] (2026), see Supplemental Material for the canonical block construction, transported-dipole character projection, full-Hilbert counterterm, thermodynamic identities, geometry, and benchmark provenance.